A Recurrent Depth-Vision Local Planner for a Wheeled Humanoid

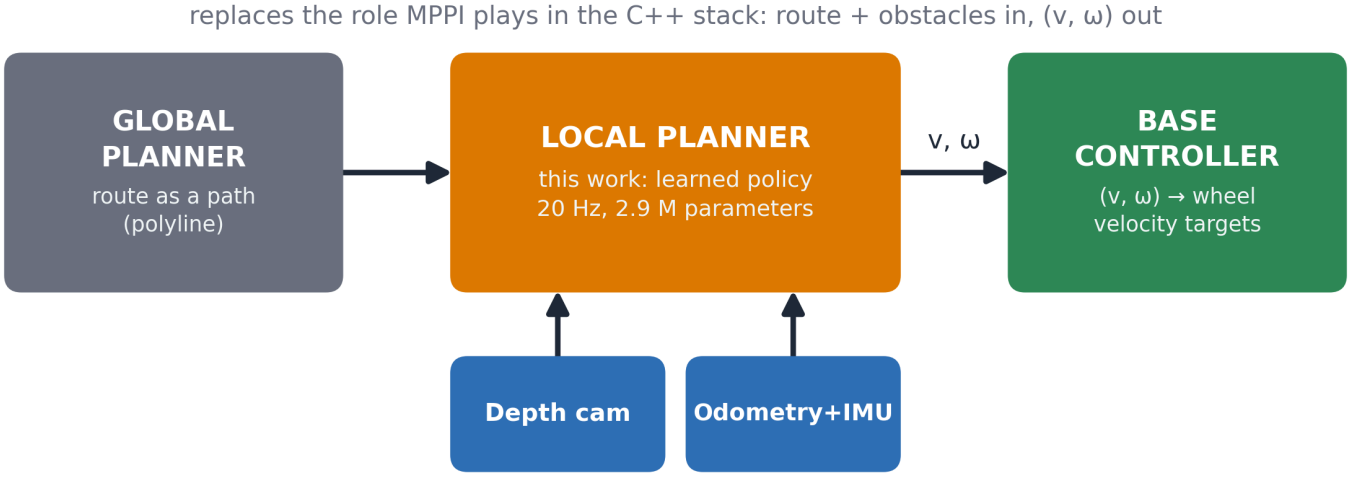
Checkpoint `MEM_GRU_model_6999` · Dynamoid v1 · Isaac Lab / rsl_rl · all results from simulation · 2026-10-11

Abstract

We train one neural network to act as the local planner of a wheeled humanoid. Every 50 ms it turns four depth frames, nine numbers describing the robot and its route, and a 128-number memory into a forward speed and a turn rate that follow a global path around walking pedestrians. The camera sees only a 120° cone out to 2.88 m, so pedestrians routinely drop out of view just before they reach the robot. Adding a GRU memory of those pedestrians, trained with an auxiliary "where were they?" loss, halved pedestrian collisions in simulation (22.7% → 11.3%) and raised goal arrival from 75.8% to 86.9%. The policy steers around people at full speed; it does not choose to slow down or stop. It has not been run on the robot.

1. Problem and setting

The robot is a differential-drive base (wheel radius 0.075 m, wheelbase 0.372 m, body radius 0.607 m) carrying a humanoid upper body. In the C++ stack its local planner has been MPPI; this policy takes over the same role: **route and obstacles in, (v, ω) out**.



The training world is a walled corridor with a 10 m reference path and three pedestrians (radius 0.30 m) walking toward the robot, either straight or with random speed (0–1.2 m/s) and heading (±35°). They do not react to the robot. The policy runs at 20 Hz over 120 Hz physics, 64 robots in parallel, episodes of at most 25 s. An episode ends at the goal (within 1 m), on a pedestrian hit (centres within 0.907 m), on a wall hit, or at timeout.

The sensor is the limiting factor. The Femto Bolt depth camera is 120° × 120°, returns nothing beyond 2.88 m, and is mounted flat, 0.09 m above the floor and 0.40 m ahead of the wheel axis. A pedestrian walking at about 1 m/s covers 2.88 m in under three seconds, and a turning robot sweeps them out of the cone. On the feedforward baseline (SOLID_BASE, 143 recorded hits), 99.3% of pedestrian hits involved someone who had been seen and then lost; the median time out of view at impact was 0.3 s. That is the case for memory.

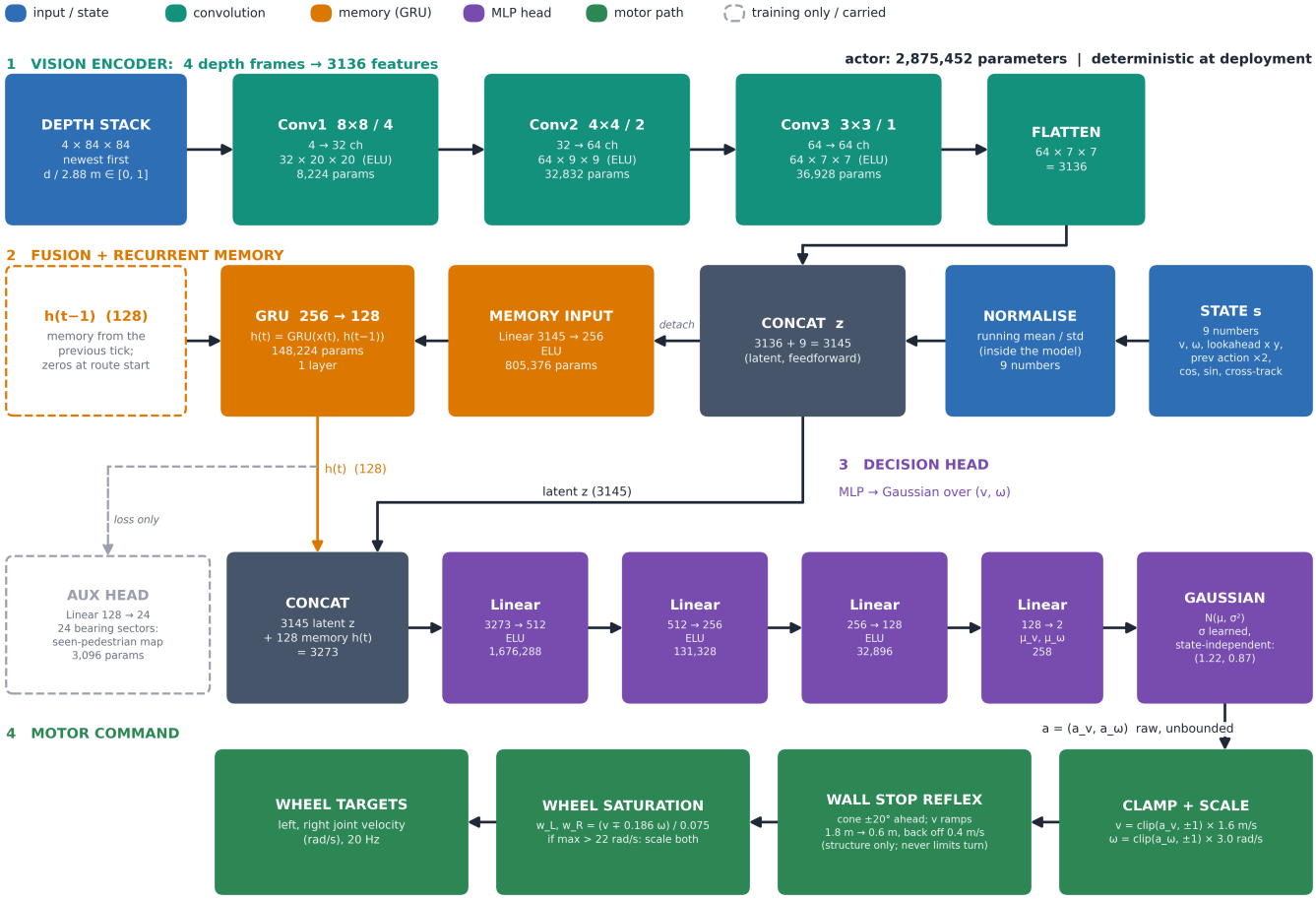
2. Method

Observations

input	shape	content
depth	4 × 84 × 84	pixel = distance / 2.88 m, invalid = 1; four consecutive frames (0.2 s), newest first
state	9	measured v, ω · path point 1.2 m ahead (x, y) · previous raw action (2) · cos, sin of path-heading error · signed cross-track error
memory	128	GRU state from the previous tick; zeros at route start

The critic, used only in training, sees 47 privileged numbers (a 24-direction time-to-collision scan, root pose and velocities, and so on). It is never deployed.

Architecture (actor, 2,875,452 parameters)



1. **Vision encoder.** Three convolutions (32@8×8/4, 64@4×4/2, 64@3×3/1, ELU) reduce the depth stack to 3,136 features. The nine state numbers, normalised inside the model, are appended: a 3,145-number latent.
2. **Memory.** The latent is detached, projected 3,145 → 256, and fed to a one-layer GRU(256 → 128). The GRU output joins the latent: 3,145 + 128 = 3,273.
3. **Decision head.** An MLP 3,273 → 512 → 256 → 128 → 2 gives the means of two Gaussians over (v, ω) with learned, state-independent widths ($\sigma_v = 1.22$, $\sigma_\omega = 0.87$). At deployment the mean is used, so the policy is deterministic given (frames, state, memory).
4. **Motor path.** $v = \text{clip}(a_v, \pm 1) \times 1.6$ m/s and $\omega = \text{clip}(a_\omega, \pm 1) \times 3.0$ rad/s. A wall reflex (±20° cone, speed ramps from 1.8 m to 0.6 m, then 0.4 m/s reverse) acts on structure only and never limits turning. Wheel speeds are then saturated at 22 rad/s.

The memory is a side branch whose columns in the head's first layer start at zero. The network therefore began identical to its memoryless parent and learned to use the memory gradually. The parameter budget is dominated by the memory input layer (805 k) and the first head layer (1.68 M); the convolutions are only 78 k.

Training

PPO (rsl_rl) with an asymmetric critic. 64 envs × 48 steps (2.4 s, which is also the memory's back-propagation window), 5 epochs × 4 minibatches, Adam with learning rate adapted to KL 0.01, clip 0.2, γ 0.995, λ 0.95, entropy 0.005. The GRU state is additionally trained to predict, for each 15° bearing slice, how close the nearest pedestrian *seen earlier this episode* was ($1 - d/4$ m; 0 if none). This is the only supervision of memory content. Each stage warm-starts the previous one **with its optimizer state**: resetting Adam alone dropped goal arrival from 75% to 8% in ten iterations.

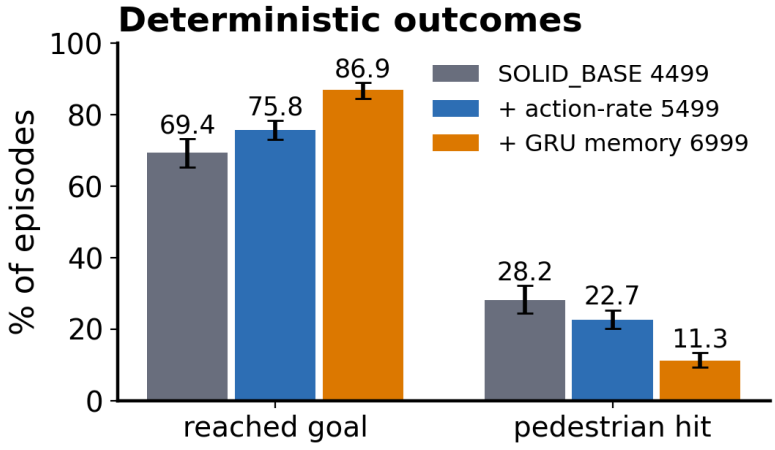
Lineage: `SOLID_BASE_model_4499` → `smooth_ar05` (action-rate penalty, 4499 → 5499) → `mem_gru128_v2` (GRU memory, 5499 → 6999).

Reward

$r = \Delta t \cdot \Sigma\, w_i\, f_i$, with $\Delta t = 0.05$ s. Terms: progress along the path (+8 per metre; +80 per 10 m), goal (+100 once), path tracking and a walking "ghost" schedule (+1 each), time-to-collision cost (−4), personal-space cost around 0.907 m (−14), action-rate penalty on clamped actions (−0.5), pedestrian hit (−200 once, episode ends), wall hit (−150), timeout (−30). Progress telescopes so it cannot be farmed, dense costs are capped so a detour never costs more than a crash, and the goal bonus sits below the collision penalty. The consequence, discussed below, is that stopping never pays.

3. Results

Deterministic policy, treadmill corridor, 3 pedestrians, 64 envs × 3000 steps; bars are 95% Wilson intervals.



n = 507 / 1038 / 929 episodes; bars: 95% Wilson interval

	SOLID_BASE 4499	+ action rate (5499)	+ GRU memory (6999)
episodes	507	1,038	929
reached goal	69.4%	75.8% (73.1–78.3)	**86.9%** (84.5–88.9)
pedestrian hit	28.2%	22.7% (20.3–25.4)	**11.3%** (9.4–13.5)
timeout	–	1.4%	1.6%
passing side L / R	–	44 / 56	44 / 56

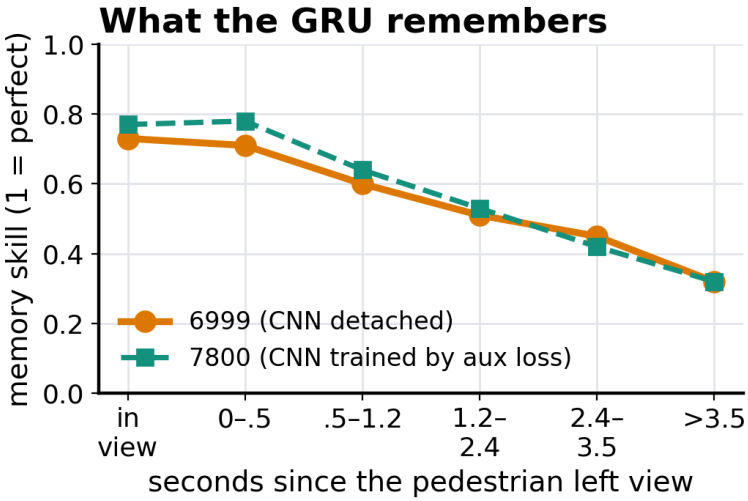
The intervals for 5499 and 6999 do not overlap. The left/right passing split is unchanged, so memory did not introduce a swerve bias. Improvement was monotone across checkpoints (6100: 19.1% hits, 6500: 15.5%, 6999: 11.3%).

Where the hits went. Per 1,000 episodes, by time the pedestrian had been out of view at impact:

out of view	5499	6999
still in view	46	6.5
≤ 0.2 s	18	7.5
0.2 – 1.2 s	118	41
1.2 – 3 s	30	39
> 3 s	3.9	10.8
total	**216**	**104**

Memory removed most of the short-blind hits it was built for (−65% at 0.2–1.2 s). In-view hits also fell 86%, plausibly because fewer pedestrians reach close range. Hits after more than 1.2 s out of view went *up* and are now 47% of what remains.

What the memory knows. Skill (1 − error / mean target) decays smoothly with time out of view.



There is no cliff at the 2.4 s training window, so a longer window is not the bottleneck. The memory is limited by what is written into it: even for pedestrians still in view, skill is only 0.73, because the CNN features feeding the GRU are detached and were trained for control, not localisation.

Forward-pass check. Re-running the checkpoint outside the simulator on the recorded observations reproduces the simulator's actions: max $|\Delta| = 0.006$ over 2,400 steps, identical clamped action on 100% of steps. This fixes the input contract a deployed node must reproduce.

4. Behaviour and limits

- **It steers but does not brake.** The raw speed output is beyond the ±1 clamp on 99.6% of steps, and within 1.8 m of a pedestrian it still requests full speed on 97% of steps. Memory improved steering, not speed control. The reward never pays for stopping.
- **Rejected experiments.** A longer memory window (not the bottleneck); training the CNN through the memory loss (10.4% hits at checkpoint 7800, within noise of 11.3%); pulling the

action mean into range (goal arrival fell from 75% to 1–28%); six discrete speed levels (stop chosen < 0.5% near pedestrians, no gain).

- **Simulation only.** Straight corridor, three non-reactive pedestrians. Crowds, reactive people and bends are untested. Real depth has lens distortion and holes the simulator lacks.
- **Not deployable yet.** It needs an ONNX export with the GRU state as input and output, a C++ node carrying that state at 20 Hz and resetting it per task, and an independent 360° lidar safety layer with a speed cap. The exporters raise on purpose until that exists.
- **Next.** A speed-aware time-to-collision cost so that slowing down can pay; then the long-blind hits.

Artifacts

All paths are under `wheeled_humanoid_manager/rl/figs/`.

what	where
checkpoint, spec, code snapshot, eval report	`checkpoints/MEM_GRU_model_6999.pt`, `.md`, `_code/`, `_eval_REPORT.md`
1080p video report (14 episodes: 11 goal, 3 pedestrian hit; sim view, the exact depth tensor, telemetry)	`video_report/MEM_GRU_model_6999/`
A0 poster, figures, network activations on recorded data	`poster/MEM_GRU_model_6999/`